

General Features

The i201 and i202 are intelligent digital I/O modules for the ionia™ system. i201 has 16 digital inputs channels whereas i202 has 8 digital input channels and 8 combined input/output channels. i202 has output read back functionality for implementing safety relevant applications.

All high current inputs (HCI) provide a wetting mechanism to overcome oxidation and a switching circuit to discharge parasitic capacitors. The HCI feature can be switched on and off by the application software. Additionally, selected inputs can be configured as frequency or counter inputs.

Every output is short circuit protected and reports directly to the central unit. Moreover, selected output channels can be configured as PWM output channels.

The configuration registers and all input/output states are accessible by the central unit through a fast and secure point-to-point connection (duagon serial star slave)

i201 and i202 have independent supply voltage feeds, which allows separately fused I/O circuits.

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overvoltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

Configuring the i201/i202 is done easily and time saving by using one of the five methods defined by IEC 61131.

The i201 and i202 are designed for harsh rolling stock environments and is fully compliant to EN 50155.

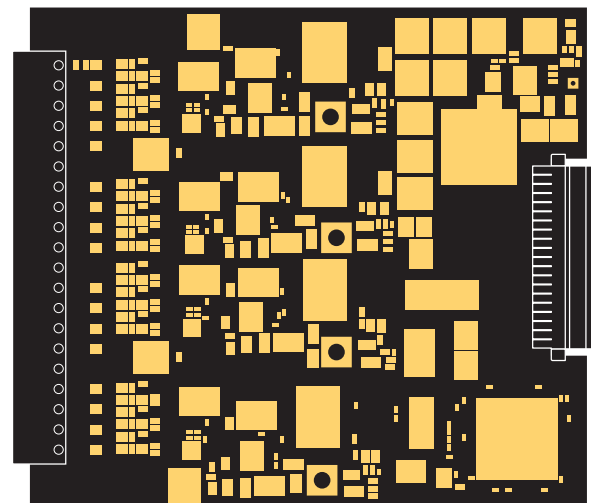
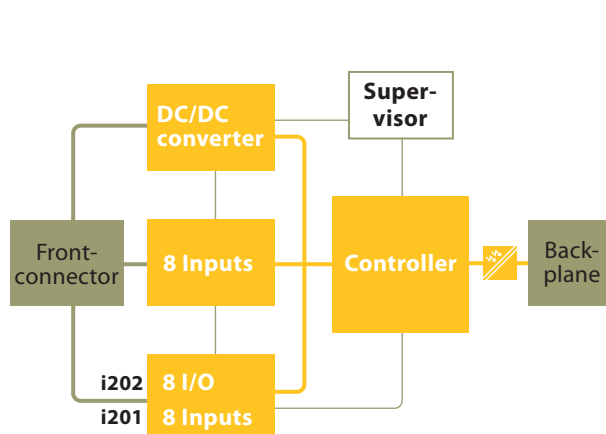
Key Benefits

- Intelligent controller features such as PWM, frequency counter, event counter, etc.
- Configurable status LEDs on each channel for easy diagnostics
- Independent supply voltage feeds, enable separately fused circuits
- «Condition Monitoring» for detailed diagnostics
- No capacitor replacement necessary
- No need for physical addressing (duagon serial star controller)
- Complies to EN 50155, EN 50121, IEC 61373

Application Examples

- General input / output controller
- Frequency counter for speed measurement or PWM input detection
- Event counters
- Analogue output functions (PWM)
- Battery monitoring and active sleep mode inputs

i201, i202 Hardware Architecture



Life Cycle Cost

The i201 and i202 have been strictly designed without electrolytic capacitor to avoid service cost for changing capacitors. The use of an FPGA with included duagon soft-IP mini-

mizes the risk and costs in case of component obsolescence. Any firmware upgrade can be done via the central unit, without the need of removing the device.

Technical Data

Input Channels	– 16 (i201) / 8 (i202) current sink input channels with wetting and switching current circuit – HCI inputs with 10 mA sink current – Nominal high level input current during scan idle time: 0.023 mA @ 24 V, 0.1 mA @ 110 V – Minimal input voltage high recognition: $0.5 \cdot V_{bat+}$ – Maximal input voltage low recognition: $0.4 \cdot V_{bat+}$ – Configurable as frequency input 10 Hz to 100 kHz	Security / Watchdog	– External supervisor for voltage, current and temperature monitoring → «Condition Monitoring» – Communication Link with watchdog supervision
Output Channels	– 8 (i202) high side (FET) from battery voltage – Read back on every output channel as option – Fail safe operation on every output channel – Overcurrent detection per output group (four Outputs) – Output short circuit protection – Nominal load current continuous: 55 W per channel, max 1 A – Short term overcurrent for 0.1 s: 6 A / Port (only one port at a time) – Nominal output switch on resistance: 0.25Ω (LV), 0.5Ω (HV) – Configurable as 16bit PWM 25 Hz to 10 kHz	Backplane Communication	– 100 Mbit duagon serial star slave – Secure communication protocol – Configurable status scan cycle time minimum: 1 ms
Supply Voltage	– 24 Vdc to 110 Vdc – Reverse polarity, overvoltage and overcurrent safe – 1000 Vac galvanic isolation – Interruption class S2 with ceramic capacitors	Diagnostic/Service	– Configurable diagnostic LEDs for each channel indicating the status of the channel or a user-defined functionality – LEDs for visual status information (power on, running, communicating, failure, I/O, ...) – Diagnostic information readable through the central unit's service interfaces (see i101/i102/i103 for further information about these possibilities)
Programming	– Programmable by the central unit using IEC 61131 LD, FBD, ST, IL and SFC – Firmware update handled by the central unit	Power Consumption	– $P_{max} < 1$ W (internal logic only) – Supports sleep mode
		Operating Conditions	– Ambient temperature: – 40 to +85 °C (EN 50 155, Class TX) – Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60 068 – Shock and vibration: According to IEC 61373 category 1, class B – EMI: According to EN 50 121 and EN 50 155
		Dimensions/Weight	– 3TE × 92 × 86 mm, fitting the ionia™ housing – 120 g
		Environment	– Fully compliant to RoHS and REACH

d-006321-013032

Order Code and HW Options

Product Type: **i** - - - EXAMPLE: i201-F or i202-N-P1-R

Config		Frequency measurement	PWM	Read back
16 inputs	i201	no on 4 channels	N F	
8 in / 8 out	i202	no on 4 channels	N F	without N with R
			no on 4 channels on 8 channels	N P1 P2

Related Documents

Data Sheet i201/i202  d-006040-nnnnnn.pdf
available at www.duagon.com

**Configure
ionia™ online:**



duagon AG
Riedstrasse 12
CH-8953 Dietikon
Phone +41 44 743 73 00
Fax +41 44 743 73 15

www.duagon.com